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import pandas as pd
import numpy as np
import pickle
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from google.colab import files

# Upload dataset during runtime
uploaded = files.upload()

file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

# Region encoding for feature consistency
if "Region" in df.columns and df["Region"].dtype == "object":
    df["Region"] = df["Region"].map({"Sidi-Bel Abbes": 0, "Bejaia": 1})

X = df.drop("FWI", axis=1)
y = df["FWI"]

# Train-test split for model robustness on unseen data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# StandardScaler for normalization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Ridge alpha tuning to improve model performance
param_grid = {"alpha": [0.01, 0.1, 1, 10, 100]}
grid_search = GridSearchCV(Ridge(), param_grid, cv=5, scoring="r2")
grid_search.fit(X_train_scaled, y_train)

best_model = grid_search.best_estimator_
y_pred = best_model.predict(X_test_scaled)

# Evaluation using MAE, RMSE and R2 Score
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("Best Alpha:", grid_search.best_params_)
print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

# Predicted vs Actual values visualization
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual FWI")
plt.ylabel("Predicted FWI")
plt.title("Predicted vs Actual FWI")
plt.grid(True)
plt.show()

# Residual analysis for error interpretation
residuals = y_test - y_pred
plt.figure(figsize=(8, 6))
plt.scatter(y_pred, residuals)
plt.axhline(y=0)
plt.xlabel("Predicted FWI")
plt.ylabel("Residuals")
plt.title("Residual Analysis")
plt.grid(True)
plt.show()

```

Choose Files cleaned_fwi...UTPUT.csv

cleaned_fwi_dataset OUTPUT.csv(text/csv) - 12869 bytes, last modified: 3/2/2026 - 100% done

Saving cleaned_fwi_dataset OUTPUT.csv to cleaned_fwi_dataset OUTPUT.csv

Best Alpha: {'alpha': 1}

MAE: 0.4887601680356688

RMSE: 0.7090874458697095

R2 Score: 0.9834103862506142

